

Relaxation

Michel Bierlaire

Practice quiz

The company Alpiq is investing in hydro-electricity. The engineers have identified four potential location to build new dams. For each location, they have assessed the investment costs as well as the long-term expected revenues:

	Location	Cost (mCHF)	Revenues (mCHF)	Return
1	China	50	160	3.20
2	Iran	70	220	3.14
3	Brazil	40	120	3
4	India	30	80	2.67

The company has a budget of 140 mCHF. Where should the company invest in new dams? This is a knapsack problem.

1. Formulate the problem as an integer optimization problem.
2. Solve the relaxation of the problem, and use the solution of the relaxation to advise the company.
3. Solve the problem by full enumeration, and use the solution to advise the company. Compare the two advises.